Recursive Skip Logic as a Human-Centered Alternative to Black-Box Adaptive Learning Systems - The Structure

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Abstract

Adaptive learning systems promise personalization but often rely on opaque algorithms, reductive data models, and engagement-based optimization. Recent legal and public scrutiny of current, widespread platforms highlights concerns regarding data privacy, behavioral profiling, and instructional inefficiency.

This paper introduces **Recursive Skip Logic (RSL)**, a framework that models learning as a dynamic, nonlinear process driven by error structure, contextual cognition, and continuous reassessment. Unlike traditional adaptive systems that rely on score-based progression and remediation loops, RSL enables multidimensional branching, conceptual reframing, and transparent decision-making.

We argue that RSL improves learning efficiency, preserves learner complexity, and addresses ethical concerns by minimizing data abstraction and restoring teacher interpretability. This work situates RSL as both a pedagogical and systemic alternative aligned with emerging critiques of algorithmic education.

Keywords

Adaptive learning, skip logic, recursive assessment, educational AI, data ethics, personalized learning, instructional systems, inclusive learning, special education, primary education, secondary education, accessibility, global learning, quality questioning, technology for future leaders, educational resources.

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1. Introduction

Adaptive learning has been marketed as the solution to differentiation, yet its implementation often collapses complexity into narrow measurable outputs. Systems like i-Ready operationalize personalization through predefined pathways, performance bands, and algorithmic adjustments that are largely opaque to educators.

Recent legal challenges and educator critiques reveal a deeper issue: these systems do not merely adapt to learners, they **model, predict, and constrain them**.

This paper proposes an alternative: **Recursive Skip Logic (RSL)**, a framework that treats learning as a dynamic system rather than a sequence of checkpoints.

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2. Problem Landscape

2.1 Reduction of Learner State

Most adaptive systems represent students using:

- correctness
- response time
- aggregate scores

This creates a lossy compression of cognition, ignoring:

- misconception types
- reasoning pathways
- contextual variability

2.2 Path Dependency and Instructional Inefficiency

Students often experience:

- repetitive remediation loops
- misplacement in difficulty bands

- limited conceptual redirection

This results in a system that reinforces error rather than transforming it.

2.3 Data and Ethical Concerns

Critiques and lawsuits highlight:

- extensive behavioral data collection
- lack of transparency
- potential misuse beyond instruction

These systems operate as **black boxes with high observational power and low accountability**.

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3. Recursive Skip Logic (RSL): The Model

RSL reframes learning as a recursive system:

Pn+1=f(Pn,En,Cn)P_{n+1} = f(P_n, E_n, C_n)Pn+1=f(Pn,En,Cn)

Where:

- PnP_nPn: learner position
- EnE_nEn: structured error patterns
- CnC_nCn: cognitive and contextual inputs

3.1 Core Principles

1. Error as Signal, Not Failure
Errors trigger:

- conceptual reframing
- alternate representations
- dimensional shifts

2. Nonlinear Pathways
Movement includes:

- forward progression
- lateral exploration
- abstraction scaling

3. Recursive Reassessment
Each interaction updates the learner model dynamically.

3.2 Skip Logic Mechanism

Instead of:

Wrong → repeat

RSL implements:

- Wrong → analyze error type
- Route to:
  - prerequisite concept
  - alternate modality
  - higher-level abstraction

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4. Comparative Analysis

Feature	**Traditional Adaptive Systems**	**Recursive Skip Logic**
Input Model	Score-based	Error-structured
Pathing	Linear/remedial	Multidimensional
Feedback	Repetition	Transformation
Transparency	Low	High

Feature	Traditional Adaptive Systems	Recursive Skip Logic
Optimization	Engagement/time	Learning efficiency

5. Ethical Architecture

RSL explicitly incorporates:

- **Data minimization**
- **Local processing preference**
- **No behavioral profiling beyond instruction**
- **Teacher-in-the-loop override**

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6. Implications

For Students

- Reduced cognitive fatigue
- Increased conceptual understanding
- Improved engagement through meaningful variation

For Teachers

- Visibility into learning pathways
- Ability to intervene and override

For Systems

- Shift from prediction → responsiveness
- Reduced reliance on large-scale behavioral data

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7. Conclusion

The convergence of legal scrutiny and pedagogical critique suggests a systemic misalignment in current adaptive learning systems. Recursive Skip Logic offers a framework that preserves learner complexity while improving instructional efficiency and transparency.

8. Limitations and Future Work

- RSL requires higher computational and instructional design complexity
- Implementation depends on teacher training and system transparency
- Empirical validation through controlled studies is required

Future research should include:

- classroom-based trials
- comparative outcome studies
- cognitive load measurement

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